Replication Plan

Simple Coplanar Waveguide Resonator Mask Targeting Metal–Substrate Interface

OSTI ID: 1983793

Auto-generated Replication Blueprint

April 8, 2026

Contents

Paper Overview

This paper presents the design and finite-element electromagnetic simulation of coplanar waveguide (CPW) resonators for superconducting quantum computing, with mask geometries specifically targeting the metal–substrate (MS) interface to study surface dielectric losses. The authors vary geometric parameters (center conductor width, gap width, film thickness) and compute electric field energy participation ratios in different loss regions (MS interface, substrate–air interface, metal–air interface, bulk substrate). The quality factors and resonance frequencies are then predicted from these participation ratios and compared with experimental measurements.

Detailed Workflow

Phase 1: Geometry Definition

1.1 Define CPW resonator geometry.

- Center conductor width w, gap s, film thickness t, substrate thickness.
- Parametric sweep values as specified in the paper.
- Ground plane extent, coupling structures (if modeled).

1.2 Define interface/surface loss layers.

- Thin dielectric layers at MS, SA, MA interfaces (thickness $\sim 3\,\mathrm{nm}$).
- Assign dielectric constants and loss tangents for each layer.

Phase 2: Finite-Element Simulation

2.1 Set up the 2D cross-section model in a FEM solver.

- Eigenmode analysis to find the CPW quasi-TEM mode.
- Adaptive mesh refinement near interfaces.

2.2 Compute electric field distribution $\mathbf{E}(x, y)$.

2.3 Compute participation ratios.

$$p_i = \frac{\int_{\mathrm{region}_i} \varepsilon_i |\mathbf{E}|^2 \, dA}{\int_{\mathrm{total}} \varepsilon |\mathbf{E}|^2 \, dA}$$

2.4 Predict quality factor.

$$\frac{1}{Q} = \sum_i p_i \tan \delta_i$$

Phase 3: Parametric Sweep

3.1 Sweep w and s over specified ranges.

3.2 For each geometry, compute participation ratios and predicted Q.

3.3 Reproduce key plots: p_{MS} vs. w/s ratio, Q vs. geometry.

Phase 4: Comparison with Experiment

4.1 Compare simulated Q values and resonance frequencies with experimental data from the paper.

4.2 Extract effective loss tangents from fitting.

Datasets

Dataset / Source	**Type**	**Location / Notes**
CPW geometry parameters	From paper	Tables of w, s, t values
Material properties	Literature	Dielectric constants and loss tangents for Si, SiO_2, Al, Nb
Experimental Q / f_0 data	From paper	Tables/figures of measured resonator properties

Models, Tools, and Software

Tool / Library	**Version**	**Purpose / URL**
COMSOL Multiphysics	6.0+	FEM electromagnetic eigenmode simulation `https://www.comsol.com/`
Ansys HFSS	latest	Alternative FEM EM solver `https://www.ansys.com/products/electronics/ansys-hfss`
Sonnet	latest	Alternative planar EM simulator `https://www.sonnetsoftware.com/`
PALACE (AWS)	latest	Open-source FEM for superconducting circuits `https://github.com/awslabs/palace`
Gmsh	≥ 4.10	Open-source mesh generation `https://gmsh.info/`
FEniCS / FEniCSx	latest	Open-source FEM framework `https://fenicsproject.org/`
Python (NumPy, Matplotlib)	≥ 3.8	Post-processing and plotting
Qiskit Metal	latest	Optional: superconducting qubit design `https://qiskit.org/metal/`

Hardware Requirements

- 2D cross-section FEM: any workstation; < 1 min per geometry.

- 3D full-resonator simulations: $\geq 32\,\mathrm{GB}$ RAM, multi-core.